MMET 410 – Lab 11

Fall 2026

****Set up RSLogix 5000 software modules**

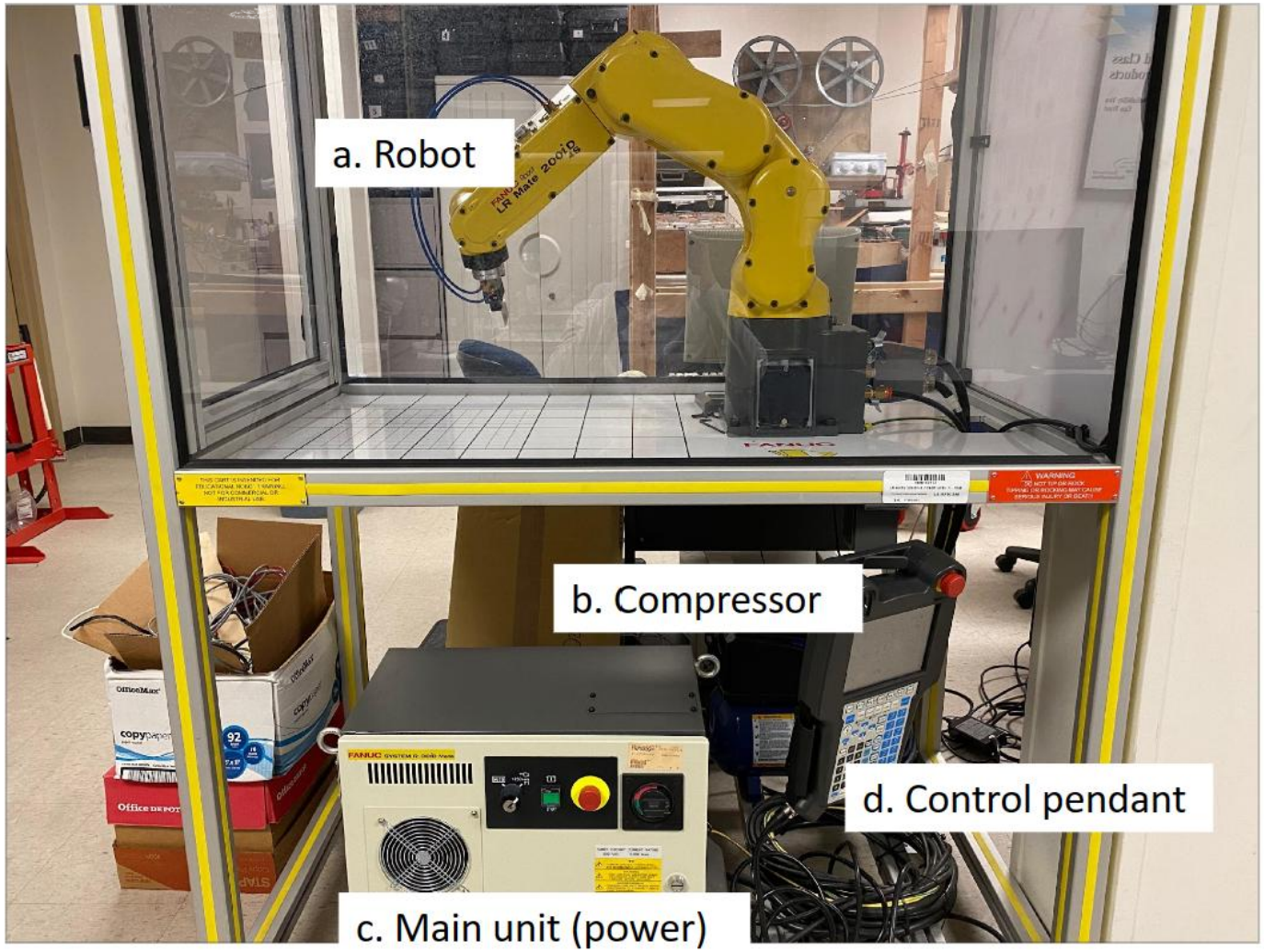
**Have the worksheet pulled out on the side!

Lab report requirements

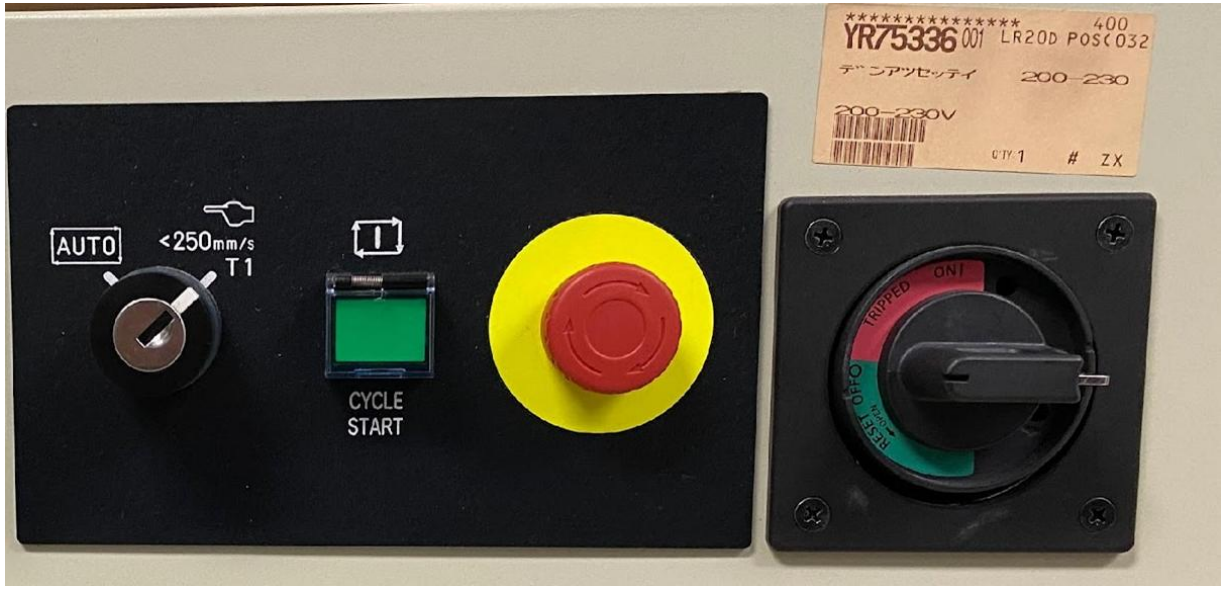
- Lab 11
- Answers to questions in task 1,2,3,4 (New added)

Report due tomorrow midnight.

Lab 11 – Task 1



Lab 11 – task 1



Q2. Pressure – 40psi

Q3. mode switch (black unit on the left)

run button (green)

emergency - stop (red)

power switch (on the right)

Q4. Two e-stop (one on the main unit, one on the pendant)

A dead-man switch on the back of the pendant

A sensor on the enclosure door

Lab 11 – task 2

1. Six Joints
2. Joint Coordinate: the current position of each joint, for more precise control.
3. World Coordinate: Cartesian coordinate of gripper, (X, Y, Z, x, y, z), x,y,z are pitch, yaw, and roll angles (orientation)
4. Shift + [+%] /[-%]



Lab 11 – task 3

- Q a. Shift+Use the ‘touch up’ button
- Q b. Highlight the line, and write the new command, and enter
- Q c. Highlight the speed part (%), then input the new speed number, and enter



Lab 11 – task 3

- Write a new program of your interest, must include the following:
  - At least 3 move commands (J P[1] 100% FINE) CNT, 10%
  - At least one WAIT commands (Wait 2sec)
  - At least one gripper control commands (RO[3]=on/off)
  - At least one JMP/LBL

Lab 11 – task 3

- Write a new program of your interest, must include the following:
  - At least 3 move commands
  - At least one WAIT commands
  - At least one gripper control commands
  - At least one JMP/LBL

Example:

LBL[1] //enter repeat label[1]

J P[1] 30% FINE //move to position 1 at 30% speed CNT100

WAIT 2 sec //hold for 2s

RO[3] = on //gripper on/open

J P[2] 100% FINE //move to position 2 at full speed

RO[3] = off //gripper off/close

JMP LBL[1] //back to repeat label[1] and repeat

Steps on the teach pendant

1. **Create the program:** Press SELECT → F2 [CREATE], type a name (use the function keys to switch upper/lower case), then press ENTER. If the DETAIL screen opens, press F3 [EDIT].
2. **Add LBL[1]:** Press NEXT, then F1 [INST] → JMP/LBL → LBL[ ]. Type 1 and press ENTER.
3. **Record motion points:**
    1. Hold the deadman switch and jog the robot to the first position. Press F1 [POINT] and pick a J or L template. This creates J P[1] 100% FINE.
    2. Move the cursor to 100%, type 30, and press ENTER.
    3. With the cursor on FINE, press F4 [CHOICE] if you want CNT instead. FINE stops exactly at the point; CNT passes through it smoothly without stopping.
    4. For each later point, jog there and press SHIFT + F1 [POINT]. This reuses the last template.
4. **Gripper command:** F1 [INST] → I/O → RO[ ]=.... Type 3, then choose ON or OFF.
5. **WAIT:** F1 [INST] → WAIT → WAIT ...(sec). Type 1 or 2.
6. **JMP LBL[1]:** F1 [INST] → JMP/LBL → JMP LBL[ ]. Type 1.

Testing

1. Switch the controller to T1 mode and lower the speed override to about 10%.
2. Press STEP to turn on single-step mode.
3. Put the cursor on line 1. Hold the deadman switch and press SHIFT + FWD to run one line at a time.
4. Once the path looks good, turn off step mode and run it continuously. The JMP makes it loop forever, so press HOLD or release the deadman switch to stop it.

Lab 11 – task 4

Problem 1:
Continuous Packaging Loop (Simple Pick and Place)

- **Scenario:** A robot must pick small blue components from a pick tray **[P1/P2]** and place them into a packing box**[P3/P4]**.

Problem 2:
Two-Station Inspection Routine (Quality Control)

- **Scenario:** A robot picks parts from the **LOADING DOCK (P[1])**, moves them to the **INSPECTION STATION (P[2])** for quality checking (**3 seconds**), then places parts on the **DROP STATION (P[3])**

